NEGATIVE TYPE GAPS FOR THETA GRAPHS AND 3-SUBDIVISIONS OF GRAPHS WITH A $K_{2,3}$-MINOR

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ABSTRACT. This paper studies two quantitative obstructions to negative type in graph metrics. Campbell, Hendrey, Lund, and Tompkins characterized metric graphs of negative type by excluding theta submetrics, asked for the sharp normalized negative type gap among theta-containing metric graphs, and conjectured a subdivision version for graphs with a $K_{2,3}$ minor. We prove the sharp gap formula

$$\Gamma_{\mathrm{inf}} = \frac{1}{24}$$

when every reduced edge segment has length at least one, using the ordered double sum

$$\gamma_d(\omega) = \sum_{x,y} \omega(x)\omega(y)d(x,y).$$

The extremal example is the theta with three unit segments. We also show that, whenever a finite graph has a $K_{2,3}$ minor, the vertex shortest-path metric of its 3-subdivision violates negative type. The argument produces an explicit six-vertex weighting with value at least 1/18, thereby proving the subdivision conjecture.

1. INTRODUCTION

All graphs in this paper are finite and simple unless explicitly stated otherwise. Graph distances are shortest-path distances.

For a metric space (M, d) and a finitely supported real function ω on M, write $X = \mathrm{supp}(\omega)$ and set

$$\gamma_d(\omega) = \sum_{x,y\in X} \omega(x)\omega(y)d(x,y).$$

We say that (M, d) has negative type when

$$\gamma_d(\omega) \le 0$$

for all such ω satisfying $\sum_{x\in X} \omega(x) = 0$. In Schoenberg's formulation, this is equivalent to the Hilbert-space embeddability of $(M, \sqrt{d})$ [6]. In the finite case it is one of the standard conditions connecting Euclidean embeddings, ℓ_1 metrics, and the inertia of the distance matrix; see Deza and Laurent [3, Chapter 6]. Thus negative type is a useful test for how graph-like metrics behave with respect to Hilbertian and cut-metric representations.

Metric graphs provide a natural continuous model for finite networks; see, for example, the survey of Bandelt and Chepoi [1] for background on graph metrics and metric graph theory. In this paper a metric graph is obtained by gluing finitely many metric segments along endpoints and then taking the induced shortest-path metric. We use a fixed finite segment decomposition when it is part of the data. When no decomposition is specified, the metric graph is understood in its reduced model, in which the edge segments are the maximal arcs between points of topological degree different from two. This convention removes the ambiguity caused by artificially subdividing a metric segment. The exact

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gap result below is unchanged if one starts with any finer decomposition whose segments all have length at least one, because the corresponding reduced segments have length at least one as well.

A theta is a metric graph consisting of two branch vertices joined by three internally disjoint metric segments. A metric graph is theta-containing if it contains an isometric copy of a theta, and theta-free otherwise.

The role of theta submetrics is explained by a theorem of Gupta, Newman, Rabinovich, and Sinclair [5]: a finite graph has the property that every nonnegative edge weighting gives an ℓ_1-embeddable shortest-path metric on the vertices if and only if the graph has no $K_{2,3}$ minor. Campbell, Hendrey, Lund, and Tompkins used this result to show that theta-free metric graphs have negative type, and then proved the converse: every theta-containing metric graph fails to have negative type [2]. Their work leaves two quantitative problems. First, they asked for the exact scale-normalized negative type gap of theta-containing metric graphs with all edge lengths at least one [2, Question 6]. Second, they conjectured that a 3-subdivision already destroys negative type for every graph containing a $K_{2,3}$ minor [2, Conjecture 9].

The purpose of this paper is to settle both problems. The first result gives the exact answer to the gap question. For a metric space (M, d), define

$$\Gamma(M, d) = \sup_{\omega} \gamma_d(\omega),$$

where the supremum is over all finitely supported $\omega : M \to \mathbb{R}$ satisfying

$$\sum_{x \in M} \omega(x) = 0, \qquad \sum_{x \in M} |\omega(x)| = 1.$$

This is the negative type gap of (M, d), a quantitative version of negative type; the terminology goes back to Doust and Weston [4]. For a family $\mathcal{M}$ of metric spaces, let

$$\Gamma_{\inf}(\mathcal{M}) = \inf_{(M,d) \in \mathcal{M}} \Gamma(M, d).$$

The normalization by the smallest edge length is essential, since multiplying all distances by $t > 0$ multiplies Γ by t.

Theorem 1. *Let $\mathcal{G}$ be the family of theta-containing metric graphs in which every reduced edge segment has length at least* 1*. Then, with the ordered-sum definition of γ_d above,*

$$\Gamma_{\inf}(\mathcal{G}) = \frac{1}{24}.$$

The same value is obtained for the subfamily of metric graphs given with an arbitrary segment decomposition whose segment lengths are all at least 1*.*

The value $1/24$ is sharp: it is attained by the theta whose three segments all have length 1. This improves the earlier general lower bound of Campbell, Hendrey, Lund, and Tompkins and identifies the true quantitative obstruction [2, Corollary 5]. It also shows that the smallest possible failure of negative type is already present in the most symmetric theta.

The second result proves the subdivision conjecture. Graphs considered with shortest-path metrics are finite and connected, so their vertex metrics are finite. If a disconnected graph contains such a minor in one component, the same argument applies to that component.

Theorem 2. *Let G be a connected graph containing $K_{2,3}$ as a minor. Let $G^{(3)}$ be the 3-subdivision of G, that is, each edge of G is replaced by a path with three internal vertices.*

Then the shortest-path metric on $V(G^{(3)})$ does not have negative type. More precisely, there is a finitely supported function $\omega : V(G^{(3)}) \to \mathbb{R}$ such that

$$\sum_v \omega(v) = 0, \qquad \sum_v |\omega(v)| = 1, \qquad \gamma_d(\omega) \geq \frac{1}{18}.$$

Theorem 2 is the graph-theoretic part of the paper. It says that a $K_{2,3}$ minor is already detected by negative type after three subdivisions of each edge. The proof adapts the minimal-theta shortcut argument and the four-case comparison used by Campbell, Hendrey, Lund, and Tompkins in the proof of their main theorem [2, Lemmas 10 and 11, proof of Theorem 3]. The new point is that in the 3-subdivision setting the opposite points can be chosen at integer distances $0, 1, 2$ from a branch vertex, so no rounding argument is needed and all witness points are actual vertices of the subdivision. The example in [2, Proposition 8], where the 2-subdivision of K_4 is ℓ_1-embeddable, shows that subdivision alone does not force failure of negative type; it is not used here as a sharpness statement under the $K_{2,3}$-minor hypothesis.

Corollary 3. *Conjecture 9 of Campbell, Hendrey, Lund, and Tompkins [2] is true.*

Proof. This is exactly Theorem 2, because a 3-subdivision replaces each edge by a path with three internal vertices. □

2. The exact negative type gap

For $a, b, c > 0$, let $\Theta(a, b, c)$ denote the theta metric graph with branch vertices u, v and three uv paths P_1, P_2, P_3 of lengths a, b, c, respectively. We parametrize P_i by arclength from u, so that u has coordinate 0 and v has coordinate a, b, c on P_1, P_2, P_3, respectively.

We first record the sharp lower bound for a single theta.

Proposition 4. *Let $1 \leq a \leq b \leq c$. Then*

$$\Gamma(\Theta(a, b, c)) \geq \frac{a}{24} \geq \frac{1}{24}.$$

Proof. Write $L_1 = a$, $L_2 = b$, and $L_3 = c$. Let m_i be the midpoint of P_i, so that m_i has coordinate $L_i/2$ on P_i.

We define a signed weighting. Start with weight $1/4$ at each of u and v, and weight $-1/6$ at each of m_1, m_2, m_3. If $b > a$, reduce the weight at v by $1/6$ and put weight $1/6$ at the point $q_2 \in P_2$ with coordinate $(b + a)/2$. If $c > a$, reduce the weight at u by $1/6$ and put weight $1/6$ at the point $q_3 \in P_3$ with coordinate $(c - a)/2$. All other weights are zero.

The total weight is zero. The positive weights have total mass $1/2$, the negative weights have total mass $-1/2$, and hence $\sum_x |\omega(x)| = 1$. It remains only to compute $\gamma_d(\omega)$.

When $a < b \leq c$, the nonzero weights are

x	u	v	q_2	q_3	m_1	m_2	m_3
$\omega(x)$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{6}$	$\frac{1}{6}$	$-\frac{1}{6}$	$-\frac{1}{6}$	$-\frac{1}{6}.$

The relevant distances are as follows:

d	u	v	q_2	q_3	m_1	m_2	m_3
u	0	a	$\frac{a+b}{2}$	$\frac{c-a}{2}$	$\frac{a}{2}$	$\frac{b}{2}$	$\frac{c}{2}$
v		0	$\frac{b-a}{2}$	$\frac{a+c}{2}$	$\frac{a}{2}$	$\frac{b}{2}$	$\frac{c}{2}$
q_2			0	$\frac{b+c}{2}$	$\frac{b}{2}$	$\frac{a}{2}$	$\frac{b+c-a}{2}$
q_3				0	$\frac{c}{2}$	$\frac{b+c-a}{2}$	$\frac{a}{2}$
m_1					0	$\frac{a+b}{2}$	$\frac{a+c}{2}$
m_2						0	$\frac{b+c}{2}$
m_3							$0.$

Substituting these values in the ordered sum gives

$$\gamma_d(\omega) = 2\sum_{x<y} \omega(x)\omega(y)d(x,y) = \frac{a}{24}.$$

If $b = a < c$, the same computation is obtained by omitting q_2 and adding its weight back to v. If $a = b = c$, it reduces to the five-point weighting

$$\omega(u) = \omega(v) = \frac{1}{4}, \qquad \omega(m_1) = \omega(m_2) = \omega(m_3) = -\frac{1}{6},$$

which again gives $\gamma_d(\omega) = a/24$. This proves the proposition. $\square$

It remains to show that the preceding lower bound is sharp. The extremal space is the unit theta $\Theta(1,1,1)$.

Lemma 5. *For the unit theta* $\Theta(1,1,1)$,

$$\Gamma(\Theta(1,1,1)) = \frac{1}{24}.$$

Proof. The lower bound follows from the weighting in the last paragraph of the proof of Proposition 4. We prove the reverse inequality.

Represent the three unit paths as three copies of $[0,1]$, with all points $(i,0)$ identified as u and all points $(i,1)$ identified as v. If $s,t \in [0,1]$, then the distance between two points on the same path is

$$A(s,t) = |s-t|,$$

and the distance between two points on distinct paths is

$$B(s,t) = 1 - |s+t-1|.$$

Let ω be an arbitrary finitely supported signed measure on the unit theta with total mass zero and total variation at most 1. Split any mass at u and v equally among the three copies of $[0,1]$. This does not change any distance contribution, because the three copies of 0 represent the same point u and the three copies of 1 represent the same point v; it also leaves the total variation unchanged. Let μ_i be the resulting signed measure on the ith copy. Put

$$\bar{\mu} = \frac{\mu_1+\mu_2+\mu_3}{3}, \qquad \nu_i = \mu_i - \bar{\mu}.$$

Then $\sum_i \nu_i = 0$, the measure $\bar{\mu}$ has total mass zero, and $\|3\bar{\mu}\| \le 1$. A direct decomposition of the path-index matrix gives

$$\begin{aligned}\gamma_d(\omega) = 3 &\iint (A+2B)(s,t)\, d\bar{\mu}(s)d\bar{\mu}(t) \\ &+ \sum_{i=1}^{3} \iint (A-B)(s,t)\, d\nu_i(s)d\nu_i(t).\end{aligned}$$

For $s,t \in [0,1]$,

$$(A-B)(s,t) = |s-t| - 1 + |s+t-1| = -2\min\{s, 1-s, t, 1-t\}.$$

The last kernel is negative semidefinite, since

$$\min\{s,1-s,t,1-t\} = \int_0^{1/2} \mathbf{1}_{[r,1-r]}(s)\mathbf{1}_{[r,1-r]}(t)\, dr.$$

Therefore the second term is nonpositive.

Set $\eta = 3\bar{\mu}$. Then η has total mass zero and $\|\eta\| \le 1$, and it suffices to prove

$$\frac{1}{3}\iint (A+2B)(s,t)\, d\eta(s)d\eta(t) \le \frac{1}{24}.$$

Let $R\eta$ be the reflection of η under $t \mapsto 1-t$, and write

$$\eta_s = \frac{\eta + R\eta}{2}, \qquad \eta_a = \frac{\eta - R\eta}{2}.$$

For zero-mass signed measures α, β on $[0,1]$, put

$$I(\alpha,\beta) = \iint |s-t| \, d\alpha(s) d\beta(t).$$

The interval metric has negative type, so $I(\alpha,\alpha) \le 0$. Since the constant part of B vanishes against zero-mass measures, and since $B(s,t) = 1 - |s-(1-t)|$, we have

$$\iint (A+2B)\, d\eta d\eta = I(\eta,\eta) - 2I(\eta, R\eta) = -I(\eta_s,\eta_s) + 3I(\eta_a,\eta_a) \le -I(\eta_s,\eta_s).$$

Here the mixed term $I(\eta_s,\eta_a)$ is zero by reflection symmetry. It remains to bound the last term. Let $F(x) = \eta_s([0,x])$. For zero-mass signed measures on an interval,

$$-I(\eta_s,\eta_s) = 2\int_0^1 F(x)^2 \, dx.$$

The measure η_s is invariant under $t \mapsto 1-t$. If $0 \le x \le 1/2$, then the two disjoint end intervals $[0,x]$ and $[1-x,1]$ have the same η_s-mass $F(x)$, while the middle interval has mass $-2F(x)$. Since $\|\eta_s\| \le \|\eta\| \le 1$, this gives $4|F(x)| \le 1$. For $x \ge 1/2$, the identity $F(x) = -F(1-x)$ follows from reflection invariance and total mass zero, so again $|F(x)| \le 1/4$. Consequently

$$-I(\eta_s,\eta_s) \le 2\int_0^1 \frac{1}{16}\, dx = \frac{1}{8}.$$

Thus

$$\gamma_d(\omega) \le \frac{1}{3}\cdot\frac{1}{8} = \frac{1}{24}.$$

Taking the supremum over all admissible ω proves the lemma. ☐

Proof of Theorem 1. We use the elementary monotonicity of the gap under isometric inclusion: if Y is an isometric subspace of X, then $\Gamma(X) \ge \Gamma(Y)$, because every admissible weighting on Y can be extended by zero to X.

Let $G \in \mathcal{G}$. Since G is theta-containing, it contains an isometric copy of some $\Theta(a,b,c)$. In the reduced model the branch vertices of this theta are vertices of G, since an interior point of a reduced edge segment has local degree two and cannot be a theta branch point. Each of the three branch-to-branch paths therefore contains at least one reduced edge segment of G; hence $a, b, c \ge 1$ after relabeling. By Proposition 4 and monotonicity,

$$\Gamma(G) \ge \Gamma(\Theta(a,b,c)) \ge \frac{1}{24}.$$

Therefore $\Gamma_{\mathrm{inf}}(\mathcal{G}) \ge 1/24$. On the other hand, the unit theta belongs to $\mathcal{G}$, and Lemma 5 gives $\Gamma(\Theta(1,1,1)) = 1/24$. Hence $\Gamma_{\mathrm{inf}}(\mathcal{G}) = 1/24$.

If the metric graph is given with a finer segment decomposition whose segment lengths are all at least 1, then each reduced edge segment is a union of such segments and has length at least 1. Thus the same proof also gives the asserted value for that subfamily. ☐

3. The 3-subdivision theorem

We now prove Theorem 2. The argument is modeled on the minimal-theta method of Campbell, Hendrey, Lund, and Tompkins [2, Lemmas 10 and 11, proof of Theorem 3], but the subdivision length allows the witness points to be chosen directly among the vertices of the subdivided graph.

Let $H = G^{(3)}$ and give H its unweighted shortest-path metric. Suppress every maximal induced path of H whose internal vertices have degree two, retaining the path length as the length of the corresponding segment. The resulting metric graph is denoted by M. Distances between vertices of H are unchanged by this compression. Every segment of M has length a positive multiple of 4.

Lemma 6. *The metric graph M contains a theta.*

Proof. Since $K_{2,3}$ is subcubic, a graph having a $K_{2,3}$ minor contains a subdivision of $K_{2,3}$. For completeness we recall the standard argument. Choose a minor model with branch sets A_1, A_2 corresponding to the degree-three vertices of $K_{2,3}$ and branch sets B_1, B_2, B_3 corresponding to the degree-two vertices. The branch sets are pairwise disjoint and connected, and each pair A_i, B_j is joined by at least one edge.

For each A_i, take a minimal tree spanning the three attachment vertices leading to B_1, B_2, B_3. This tree has a center a_i from which three internally disjoint subpaths reach the three attachments, allowing one of these subpaths to have length zero. For each B_j, take a path inside B_j between its two attachments. Since the branch sets are disjoint, these choices give three internally disjoint a_1a_2 paths. Hence the underlying graph contains a theta subgraph. Subdividing every edge three times and then suppressing degree-two chains preserves this theta as a theta subgraph of M. □

Choose, among all theta subgraphs of M, one of minimum total length; call it T. Let u and v be its branch vertices, and write its three internally disjoint uv paths as P_1, P_2, P_3, ordered by length:

$$|P_1| \leq |P_2| \leq |P_3|.$$

The intrinsic distance in T is denoted by d_T.

Lemma 7. *Let S be a path in M whose endpoints p, q lie on T, whose interior is disjoint from T, and such that no one of the paths P_1, P_2, P_3 contains both p and q. Then*

$$|S| \geq \max\{d_T(u,p), d_T(v,p), d_T(u,q), d_T(v,q)\}.$$

Proof. Let $p \in P_a$ and $q \in P_b$, where $a \neq b$, and let P_c be the remaining uv path. Put

$$p_u = d_T(u,p), \quad p_v = d_T(v,p), \quad q_u = d_T(u,q), \quad q_v = d_T(v,q).$$

The union $T \cup S$ contains several theta subgraphs. Since T was chosen with minimum total length, each of them has total length at least $|P_a| + |P_b| + |P_c|$.

First consider the theta with branch vertices u and q whose three internally disjoint paths are

$$P_b[u,q], \qquad P_a[u,p] \cup S, \qquad P_c \cup P_b[v,q].$$

Its total length is

$$q_u + (p_u + |S|) + (|P_c| + q_v) = |P_b| + |P_c| + p_u + |S|.$$

Minimality gives $|S| \geq |P_a| - p_u = p_v$. Interchanging u and v gives $|S| \geq p_u$.

Next consider the theta with branch vertices u and p whose three internally disjoint paths are

$$P_a[u,p], \qquad P_b[u,q] \cup S, \qquad P_c \cup P_a[v,p].$$

Its total length is

$$p_u + (q_u + |S|) + (|P_c| + p_v) = |P_a| + |P_c| + q_u + |S|.$$

Minimality gives $|S| \geq |P_b| - q_u = q_v$. Interchanging u and v gives $|S| \geq q_u$. Combining the four inequalities proves the claim. □

Lemma 8. *If $x \in T$ and $d_T(u,x) \leq 2$, then*

$$d_M(x,y) = d_T(x,y) \qquad \textit{for every } y \in T.$$

The analogous statement with v in place of u is also true.

Proof. We prove the assertion at u; the proof at v is identical. Assume that some $y_0 \in T$ satisfies $d_M(x, y_0) < d_T(x, y_0)$, and take a shortest xy_0 path Q in M. Walking from x along Q, let y be the first point of $Q \cap T$ for which the initial subpath from x to y is shorter than $d_T(x, y)$. Let z be the last point of T before this y on that initial subpath. Then the part from x to z is intrinsic in T, while the subsequent zy subpath S has interior disjoint from T and satisfies

$$d_T(x, z) + |S| < d_T(x, y).$$

In particular $|S| < d_T(z, y)$.

If one of the paths P_1, P_2, P_3 contains both z and y, then replacing the zy subpath of that path by S gives a theta shorter than T, a contradiction. Hence no P_i contains both endpoints. Lemma 7 gives $|S| \geq d_T(u, y)$.

A path can leave T only at a vertex of the compressed graph M. Thus, if $z \neq u$, the point z is a compressed vertex of T different from u. Since each segment of M has length at least 4, this yields $d_T(u, z) \geq 4$. Together with $d_T(u, x) \leq 2$ it follows that $d_T(x, z) \geq d_T(x, u)$; when $z = u$ this inequality is immediate. Therefore

$$d_T(x, y) \leq d_T(x, u) + d_T(u, y) \leq d_T(x, z) + |S|,$$

contradicting the strict inequality above. No shorter ambient path exists. □

For $t = 0, 1, 2$, let p_t be the vertex of H lying on P_1 at distance t from u. For $j = 2, 3$, let $\psi_j(p_t)$ be the point antipodal to p_t in the cycle $P_1 \cup P_j$, and define

$$r_t = \psi_2(p_t), \qquad s_t = \psi_3(p_t).$$

Put

$$C_j = \frac{|P_1| + |P_j|}{2} \qquad (j = 2, 3).$$

Then r_t lies on P_2 at distance $C_2 - t$ from u, and s_t lies on P_3 at distance $C_3 - t$ from u. Since all three path lengths are multiples of 4, both C_2 and C_3 are even integers. Thus all points p_t, r_t, s_t with $t = 0, 1, 2$ are vertices of H. Also, the open intervals from r_0 to r_2 and from s_0 to s_2 contain no compressed vertex of M.

For $i = 0, 1$, Lemma 8 gives ambient distances from p_i and p_{i+1} to points of T. In each cycle $P_1 \cup P_j$, the antipodal pairs have distance C_j, while the crossed pairs have distance $C_j - 1$. Hence

$$\begin{aligned} &d(p_i, r_i) + d(p_{i+1}, r_{i+1}) + d(p_i, s_i) + d(p_{i+1}, s_{i+1}) \\ &\quad = d(p_i, r_{i+1}) + d(p_{i+1}, r_i) + d(p_i, s_{i+1}) + d(p_{i+1}, s_i) + 4. \end{aligned} \tag{1}$$

The next lemma supplies the only comparison not forced by the antipodal construction.

Lemma 9. *If the intervals $[r_0, r_2]$ and $[s_0, s_2]$ are disjoint, then some $i \in \{0, 1\}$ satisfies*

$$d(r_i, s_i) + d(r_{i+1}, s_{i+1}) \geq d(r_i, s_{i+1}) + d(r_{i+1}, s_i).$$

Proof. A shortest path from r_1 to s_1 must leave the first interval through r_0 or r_2 and must enter the second interval through s_0 or s_2, because the interiors of these intervals have no compressed vertices. Consequently

$$d(r_1, s_1) = 2 + \min\{d(r_0, s_0), d(r_0, s_2), d(r_2, s_0), d(r_2, s_2)\}.$$

We check the four possible minimizers.

If the minimum is $d(r_0, s_0)$, then

$$\begin{aligned} d(r_0, s_1) + d(r_1, s_0) &\leq (d(r_0, s_0) + 1) + (1 + d(r_0, s_0)) \\ &= d(r_0, s_0) + d(r_1, s_1), \end{aligned}$$

which gives the desired inequality with $i = 0$. If the minimum is $d(r_2, s_2)$, the same estimate at the other end gives the desired inequality with $i = 1$.

If the minimum is $d(r_0, s_2)$, then

$$\begin{aligned} d(r_0, s_1) + d(r_1, s_0) &\le (d(r_0, s_2) + 1) + (d(r_0, s_0) + 1) \\ &= d(r_1, s_1) + d(r_0, s_0), \end{aligned}$$

so $i = 0$ works. Finally, if the minimum is $d(r_2, s_0)$, then

$$\begin{aligned} d(r_0, s_1) + d(r_1, s_0) &\le (d(r_0, s_0) + 1) + (d(r_2, s_0) + 1) \\ &= d(r_0, s_0) + d(r_1, s_1), \end{aligned}$$

and again $i = 0$ works. □

If $|P_1| = |P_2| = |P_3|$, then $r_0 = s_0 = v$ and Lemma 9 is not used. In this exceptional case choose $i = 1$. The four points r_1, r_2, s_1, s_2 are within distance 2 of v, so Lemma 8 applied at v reduces their pairwise distances to intrinsic distances in the theta. Hence

$$d(r_1, s_1) + d(r_2, s_2) = 2 + 4 = 3 + 3 = d(r_1, s_2) + d(r_2, s_1).$$

Combining this exceptional case with Lemma 9, we may choose $i \in \{0, 1\}$ so that

$$d(r_i, s_i) + d(r_{i+1}, s_{i+1}) \ge d(r_i, s_{i+1}) + d(r_{i+1}, s_i), \tag{2}$$

and the six points below are distinct vertices of H.

Set

$$B = \{p_i, r_i, s_i\}, \qquad R = \{p_{i+1}, r_{i+1}, s_{i+1}\}.$$

Define

$$\Delta = \sum_{\{r,r'\}\subset R} d(r, r') + \sum_{\{b,b'\}\subset B} d(b, b') - \sum_{r\in R,\, b\in B} d(r, b).$$

Equations (1) and (2) show that the six within-set terms involving points on different theta paths exceed the corresponding six cross terms by at least 4. The remaining cross terms are the three unit distances

$$d(p_i, p_{i+1}) = d(r_i, r_{i+1}) = d(s_i, s_{i+1}) = 1.$$

Thus

$$\Delta \ge 4 - 3 = 1. \tag{3}$$

Finally put

$$\omega(r) = \frac{1}{6} \quad (r \in R), \qquad \omega(b) = -\frac{1}{6} \quad (b \in B),$$

and set ω equal to zero at all other vertices. Then $\sum_v \omega(v) = 0$ and $\sum_v |\omega(v)| = 1$. Since γ_d is an ordered double sum,

$$\gamma_d(\omega) = \frac{\Delta}{18} \ge \frac{1}{18}.$$

This proves Theorem 2.

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Data Availability Statement

Data sharing is not applicable to this article because no datasets were generated or analyzed.

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